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Integrating SAS® Enterprise Guide with SAS® Viya®

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Abstract

This hands-on workshop shows you how to use SAS® Enterprise Guide with SAS® Viya®. You learn how to connect Enterprise Guide to your SAS® Viya® server, then run tasks and programs on the Viya® Compute server. You also learn how to convert SAS® 9 projects to use the Compute server and how to execute code on the CAS server for high-performance processing.

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Demo: Connect to a SAS Viya Server

Overview

In this demonstration, we use SAS Enterprise Guide and establish a server connection to SAS Viya. This integration enables you to continue using your SAS Enterprise Guide Windows application and offers enhanced capabilities, such as providing access to a cloud-native, high-performance analytics platform that supports multi-language coding.

Connect to the SAS Viya Compute Server

1. If necessary, on the Windows toolbar, click the icon for **SAS Enterprise Guide**.
2. On the application toolbar, click **File→New→Project**.
3. On the application toolbar, select **Tools→Connections**.
4. In the **Connections** window, select the **Servers** tab.
   Notice the default server is set to **SASApp**.
   Note: If you see an additional server has already been set up, you can leave it or delete it. I'll be deleting mine to avoid confusion but leaving it will not have any impact on the demo.
5. Click **Add→SAS Viya 4 Compute Server**.
6. Complete the following the fields:
   Name: **SAS Viya Compute**
   System URL: **https://server.demo.sas.com**
   File Navigation: **System root**
7. Click **Add** and then **Close**.
8. In the **Servers** pane, right-click the **SAS Viya Compute** server and click **Connect**.
9. Google Chrome will open. If necessary, sign in with the following credentials:
   User ID: **student**
   Password: **Metadata0**
10. Click **Authorize Access**. Copy the **Authorization Code** provided.
11. Minimize **Google Chrome** to return to **SAS Enterprise Guide**.
12. Paste the authorization code in the new window and click **OK**.

Explore the Servers

1. In the **Servers** pane, navigate **SAS Viya Compute→Libraries→SASHELP**. Right-click on the **CARS** table and select **Add to project**.
2. In the **Process Flow**, hover over the **CARS** node and confirm it resides on the **SAS Viya Compute** server.
3. In the **Servers** pane, navigate **Servers→SASApp→Libraries→SASHELP**. Right-click on the **CARS** table and select **Add to project**.

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4. In the **Process Flow**, hover over the second **CARS** node and confirm it resides on the **SASApp** server.
   Multiple servers can be used in the same process flow.
5. On the application toolbar, select **File→New→Program**.
   Notice on the program toolbar there is a drop-down to select the server the program will run on. The new program defaulted to **SASApp**.
6. On the application toolbar, select **File→Close project** and click **Don't Save**.

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Demo: Update Existing SAS Enterprise Guide Projects

Overview

When working with a project created to run on SAS 9, you may need to convert nodes or the entire project to run on the SAS Viya Compute server.

In this demonstration, we'll start by modifying individual nodes. Then we'll see an alternative method to update all nodes using the Project Maintenance Wizard.

Run a Project on the SAS 9 SASApp Server

1. If necessary, open **SAS Enterprise Guide**.
2. On the application toolbar, select **File→Open→My Computer→Browse**. Navigate to **W:\VS**, select **Exceptions.egp**, and click **Open**.
   *Note: The **W:** drive is named **workshop**.*
3. On the application toolbar, select **View→Submission Status**.
   This pane enables you to view and manage the status of submitted programs.
4. On the **Process Flow** toolbar, click **Run**.
   Check the **Submission Status** pane to confirm all three tasks ran successfully on the **SASApp** server.
5. In the **Submission Status** pane, click **Clear All** to remove all submissions.
6. In the **Servers** pane, right-click on the **SASApp** server and select **Disconnect**.
   This deletes the **QUERY1** and **QUERY2** tables from the **WORK** library.

Update Individual Project Nodes to Run on the SAS Viya Compute Server

1. In the **Process Flow**, right-click the **CARS** node and select **Properties**.
2. In the **Properties** window, on the **General** tab, click **Change**.
3. Navigate to **Servers→SAS Viya Compute→Libraries→SASHELP** and select **CARS**. Click **Open** and then **OK**.
4. On the **Process Flow** toolbar, click **Run**.
   Check the **Submission Status** pane to see that the two tasks that used **CARS** for input ran successfully on the **SAS Viya Compute** server, but the task that used **QUERY2** for input failed on the **SASApp** server.
5. In the **Submission Status** pane, click **Clear All** to remove all submissions.
6. In the **Process Flow**, right-click on the **Bar Chart** node and select **Select input data→WORK.Query2**.
7. In the **Process Flow**, right-click the unused **QUERY2** node, select **Delete**, and then click **Yes**.
8. On the **Process Flow** toolbar, click **Run**.
   Check the **Submission Status** pane to confirm all three tasks ran successfully on the **SAS Viya Compute** server.

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9. In the **Submission Status** pane, click **Clear All** to remove all submissions.
10. On the application toolbar, select **File→Close project** and click **Don't Save**.

Update an Entire Project to Run on the SAS Viya Compute Server

1. On the application toolbar, select **File→Open**. On the **Recent** tab, double-click **Exceptions.egp**.
2. On the application toolbar, select **Tools→Project maintenance** and click **Close and Continue**.
   Note: Project maintenance will overwrite an existing project. If necessary, make a copy of the project prior to running the project maintenance so you have a backup.
3. In the **Project Maintenance** window, select the **Libraries** tab.
   Notice that **SASHELP** is located on the **SASApp** server.
4. Select the **Servers** tab. Click **Replace In Project**.
5. Select **SAS Viya Compute** and click **OK**.
6. Select the **Libraries** tab.
   Confirm that **SASHELP** is now located on the **SAS Viya Compute** server.
7. Select the **Data Items** tab. Click **Verify Data Exists**.
   Notice the **Verification Status** changes from **Unknown** to **Exists**.
8. Click **OK** and **Yes**.
9. On the **Process Flow** toolbar, click **Run**.
   Check the **Submission Status** pane to confirm all three tasks ran successfully on the **SAS Viya Compute** server.
10. In the **Submission Status** pane, click **X** to close the pane.
11. On the application toolbar, select **File→Close project** and click **Don't Save**.

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Demo: Use SAS Viya Capabilities in a SAS Enterprise Guide Project

Overview

Connecting to SAS Viya from SAS Enterprise Guide enhances collaboration by integrating open-source languages and enables parallel data processing and advanced analytics through SAS Cloud Analytics Services (CAS).

In this demonstration, we'll use Pandas in Python within SAS Enterprise Guide to read raw data, CAS code to upload the data to an in-memory table, and SAS Visual Analytics to visualize the data.

Execute Python Code

1. If necessary, open **SAS Enterprise Guide**.
2. On the application toolbar, select **File→Open→My Computer→Browse**. Navigate to **W:\VS**, select **Getting Started with PROC Python.sas** and click **Open**.
   *Note: For an in-depth overview of this program code, see the **Explanation of the Getting Started with PROC PYTHON.sas** Program section.*
3. On the program toolbar, click **Run**.
   The program **Log** returns an error message that says the **PYTHON** procedure was not found. Notice the program ran on the **SASApp** server.
4. On the program toolbar, use the drop-down to change the server to **SAS Viya Compute**.
5. On the program toolbar, click **Run**.
   Confirm the program ran successfully.
6. In the **Servers** pane, navigate to **Servers→SAS Viya Compute→Libraries→WORK**.
   Confirm **HOME_EQUITY_COMPUTE_SAS** is listed.
   *Note: If you don't see the table, right-click on the **WORK** library and select **(Refresh)**.*

Load Data into CAS

1. On the Windows toolbar, click the icon for **Google Chrome**.
2. Click the **+** button to add a new tab.
3. On the browser bookmarks bar, click the shortcut for **SAS Landing**.
4. On the **SAS Viya** toolbar, click **⁝⁝⁝ (Applications menu)** and select **Manage Data**.
   This opens the **SAS Data Explorer** application.
5. On the **Sources** tab, expand **cas-shared-default**, and select **CASUSER**.
   Confirm **HOME_EQUITY_CAS_SAS** is not listed.
6. On the Windows toolbar, click the icon for **SAS Enterprise Guide**.
7. Add the following code starting on **line 38** of the program. Using the table below, modify the code as shown for each step. After each modification, highlight the appropriate portion of the code and click the **Run** button on the program toolbar. Examine the changes as outlined.

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*Note: If you don't see what's expected in the **Servers** pane, right-click on **Libraries** and select **(Refresh)**.*

1. Add a **CAS** statement to connect to the **CAS** server.

```
cas conn;
```

*Check the program **Log** to confirm the session connected successfully.*

2. Add a **LIBNAME** statement to create a SAS library that maps to the **CASUSER** caslib.

```
libname casuser cas caslib=casuser;
```

*Check the **Servers** pane and navigate to **Servers →SAS Viya Compute →Libraries** to confirm the **CASUSER** library is listed.*
*DO NOT expand the **CASUSER** library.*

3. Add a **PROC CASUTIL** step with a **LOAD** statement to create a copy of the SAS data set on the **CAS** server and promote it.

```
proc casutil;
     load data=work.home_equity_compute_sas
          casout="home_equity_cas_sas"
          outcaslib="casuser" promote;
quit;
```

*Check the **Servers** pane, navigate to **Servers →SAS Viya Compute →Libraries**, and expand the **CASUSER** library to confirm the new file is listed.*

4. No changes. Rerun the **PROC CASUTIL** step.

```
proc casutil;
     load data=work.home_equity_compute_sas
          casout="home_equity_cas_sas"
          outcaslib="casuser" promote;
quit;
```

*Check the program **Log** to see an error message indicating the table already exists.*

5. Add a **DROPTABLE** statement to delete the table before recreating it.

```
proc casutil;
     droptable casdata="home_equity_cas_sas"
               incaslib="casuser";
     load data=work.home_equity_compute_sas
          casout="home_equity_cas_sas"
          outcaslib="casuser" promote;
quit;
```

*Check the program **Log** to confirm the table was deleted and then recreated.*

6. Add an asterisk (*) to comment out the **LOAD** statement.

```
proc casutil;
     droptable casdata="home_equity_cas_sas"
               incaslib="casuser";
    *load data=work.home_equity_compute_sas
           casout="home_equity_cas_sas"
           outcaslib="casuser" promote;
quit;
```

*Check the program **Log** to confirm the table was deleted.*

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7. No changes. Rerun the **PROC CASUTIL** step.

```
proc casutil;
    droptable casdata="home_equity_cas_sas"
              incaslib="casuser";
   *load data=work.home_equity_compute_sas
         casout="home_equity_cas_sas"
         outcaslib="casuser" promote;
quit;
```

*Check the program **Log** to see an error message that the table does not exist.*

8. Add the **QUIET** option to the **DROPTABLE** statement to prevent the error message.

```
proc casutil;
    droptable casdata="home_equity_cas_sas"
              incaslib="casuser" quiet;
   *load data=work.home_equity_compute_sas
         casout="home_equity_cas_sas"
         outcaslib="casuser" promote;
quit;
```

*Check the program **Log** to confirm there is no longer an error message.*

9. Remove the asterisk (*) to uncomment out the **LOAD** statement. Add the **LIST** statement to see a report of all tables in the **CASUSER** caslib.

```
proc casutil;
    droptable casdata="home_equity_cas_sas"
              incaslib="casuser" quiet;
    load data=work.home_equity_compute_sas
         casout="home_equity_cas_sas"
         outcaslib="casuser" promote;
    list tables;
quit;
```

*Check the program **Results** to confirm the table is listed and it is promoted.*

10. Add a **CAS** statement to disconnect from the **CAS** server.

```
cas conn terminate;
```

*Check the **Servers** pane and navigate to **Servers →SAS Viya Compute →Libraries** to confirm the **CASUSER** library is no longer listed.*

8. On the Windows toolbar, click the icon for **Google Chrome**.
9. In **SAS Data Explorer**, click **(Refresh)**.
   Confirm **HOME_EQUITY_CAS_SAS** is now listed.

Explore and Visualize with SAS Visual Analytics

1. On the **SAS Viya** toolbar, click **(Applications menu)** and select **Explore and Visualize**.
   This opens the **SAS Visual Analytics** application.
2. Click **New report**.
3. Select the **Data** pane and click **Add data**.
4. In the **Choose Data** window, select **HOME_EQUITY_CAS_SAS** and click **Add**.
5. In the **Data** pane, click and drag **loanstatus** into the work area.
   Confirm a bar chart showing the number of repaid versus defaulted loans was created.

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6. Right-click the bar chart and select **Explain data→loanstatus** and click **Add full explanation to new page**.
7. On the application toolbar, click **⋮ (More)→Close** and click **Don't save**.

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Explanation of the Getting Started with PROC Python.sas Program

PROC PYTHON Step

Lines 2-3: The **PROC PYTHON** statement begins the step, followed by the **SUBMIT** statement.

Lines 6-8: The Python code is included below the **SUBMIT** statement. The program imports the **pandas** and **numpy** packages, then sets a DataFrame column display option.

Line 12: The **read_csv pandas** method reads the **home_equity.csv** file from the SAS support website and creates a DataFrame.

Lines 16-23: **pandas** is used to prepare the DataFrame by replacing missing numeric values in the numeric columns with the mean of their respective columns, replacing character values in the character columns with the mode of their respective columns, creating two new columns, and then renaming columns by converting column names to lowercase and removing the underscore.

Lines 27-28: This code uses **head** method to print the first five rows of the DataFrame and the **isna** and **sum** methods to count the number of missing values in each column.

Line 32: To process the DataFrame in either the SAS **Compute** server or the **CAS** server in SAS Viya, it must be converted to a SAS table. The **SAS.df2sd** method transfers data from the specified DataFrame, **df**, to the SAS table **home_equity_compute_sas** in the **WORK** library.

Lines 34-35: The **PYTHON** procedure ends with the **ENDSUMBMIT** and **QUIT** statements.

CAS Connection

Line 39: The **CAS** statement is used to create a CAS session named **conn**.

Line 42: To access and manipulate data that is stored in the CAS session from **SAS Enterprise Guide**, we can use the **LIBNAME** statement to map a SAS library to the **CASUSER** caslib. This is not required for uploading the data, but serves as a useful verification method to confirm that the upload is successful.

Lines 45 and line 54: Next, the **CASUTIL** procedure is used to manage and transfer data between the SAS **Compute** server and the **CAS** server.

Line 47: The **DROPTABLE** statement drops the table if it already exists on the CAS server.

Line 50: The **LOAD** statement reads the SAS table **home_equity_compute_sas** and creates an in-memory CAS table named **home_equity_cas_sas**, within the **CASUSER** caslib on the **CAS** server. The **PROMOTE** option specifies the table should have global scope, which makes the table available to all sessions that use the **CASUSER** caslib, subject to access controls.

Line 53: The **LIST TABLES** statement lists all available CAS tables so you can confirm the table was loaded.

Line 54: The **QUIT** statement ends the **CASUTIL** procedure.

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Line 57: The **CAS** statement terminates the CAS connection and de-assigns the **CASUSER** caslib.

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